

Dr.Doc MEDICAL EDUCATION SERIES

AUTONOMIC NERVOUS SYSTEM (ANS) PHYSIOLOGY & PHARMACOLOGY

High-Yield Master Note & Complete Board Review Guide

Module Subject: Human Physiology — Autonomic Nervous System (ANS)

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Key Topics Covered: Nervous System Architecture, Cranial Nerves (I-XII), Sympathetic vs. Parasympathetic Outflow, ANS Neurotransmitters & Receptor Subtypes (M1-M5, α 1- α 2, β 1- β 3), Intracellular Signal Transduction, Organ Effects Matrix, Autonomic Reflex Arcs, ANS Pharmacology, High-Yield Summary & Board MCQs.

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MEDICAL EDUCATION

The nervous system is structurally divided into the Central Nervous System (CNS), comprising the brain and spinal cord as the integration and control center, and the Peripheral Nervous System (PNS), which encompasses all nervous tissue outside the CNS (consisting of 31 pairs of spinal nerves and 12 pairs of cranial nerves that carry information to and from the spinal cord).

The efferent division of the PNS is divided into two primary functional components:

- **Somatic Nervous System (SoNS):** Innervates skeletal muscle cells. Neuronal cell bodies reside in groups within the Anterior Horn Cells (AHC) of the gray matter of the spinal cord. Their large diameter, myelinated axons pass without any synapses directly to skeletal muscle cells (intrafusal and extrafusal fibers), releasing Acetylcholine (ACh) to cause muscle contraction.

- **Autonomic Nervous System (ANS):** Controls involuntary visceral activities (cardiovascular system, gastrointestinal tract, glandular secretion). Originates from the **Lateral Horn** of the gray matter of the spinal cord and is further subdivided into the **Sympathetic Nervous System (SyNS)**, **Parasympathetic Nervous System (PaNS)**, and the intrinsic **Enteric Nervous System (ENS)**.

Table 1.1: Detailed Comparison of Somatic (SoNS) vs. Autonomic (ANS) Nervous Systems

Feature / Characteristic	Somatic Nervous System (SoNS)	Autonomic Nervous System (ANS)
Target & Functional Control	Voluntary activities (contraction of skeletal muscles).	Involuntary activities (CVS, GIT, glands, smooth/cardiac muscle).
CNS Spinal Cord Origin	Anterior Horn of Gray Matter (AHC).	Lateral Horn of Gray Matter.
Neuronal Chain & Synapses	Single long, thick, myelinated motor axon (no peripheral ganglion/synapse).	Two-neuron relay chain: Preganglionic neuron and Postganglionic neuron.
Neurotransmitters (NTs)	Exclusively Acetylcholine (ACh).	Both Acetylcholine (ACh) and Norepinephrine (NE).
Receptor Types	Exclusively Nicotinic Receptors (N _M).	Cholinergic (Nicotinic N _N , Muscarinic M1-M5) and Adrenergic (alpha-1, alpha-2, beta-1, beta-2, beta-3).
Effector Action	Always excitatory (causes contraction).	Excitatory or Inhibitory depending on target organ and receptor type.

MODULE 2: Cranial Nerves Physiology & Clinical Pathology

The 12 pairs of cranial nerves (CN I - XII) are components of the PNS designated by Roman numerals. They perform sensory, motor, and autonomic parasympathetic functions. Specifically, **3 are pure sensory** (CN I, II, VIII), **5 are pure motor** (CN IV, VI, XI, XII, V motor portion), and several are **mixed nerves**.

CN	Nerve Name	Sensory Function	Motor Function	Autonomic Parasympathetic Function
I	Olfactory	Smell	—	—
II	Optic	Vision	—	—
III	Oculomotor	—	4 Extrinsic eye muscles (Superior, Medial, Inferior Recti; Inferior Oblique) → upward/eye movement	Pupillary constriction (iris constrictor), Accommodation & Focusing (ciliary muscle)
IV	Trochlear	—	1 Extrinsic eye muscle (Superior Oblique)	—
V	Trigeminal	Somatic senses (Face, tongue), Corneal sensitivity	Chewing / Mastication (Masseter, Temporalis muscles)	—
VI	Abducens	—	1 Extrinsic eye muscle (Lateral Rectus)	—
VII	Facial	Taste	Muscles of facial expression	Salivary glands, Tear (lacrimal) glands

VIII	Auditory (Vestibulocochlear)	Hearing (Cochlear division), Equilibrium & Balance (Vestibular division)	—	—
IX	Glossopharyngeal	Taste, Blood gases monitoring	Swallowing, Gagging	Salivary glands
X	Vagus	Blood pressure, Blood gases, Taste (minor)	Speech (laryngeal muscles), Swallowing, Gagging	Major parasympathetic nerve: innervates heart, gut/intestines, lungs, kidneys, liver, stomach
XI	Spinal Accessory	—	Neck muscles (Sternocleidomastoid, Trapezius)	—
XII	Hypoglossal	—	Tongue muscles (speech & tongue movements)	—

CRITICAL EXAM HIGHLIGHT: Cranial Nerves Functional Groups & Pathologies

High-Yield Functional Groupings of Cranial Nerves:

- **5 Special Senses:** Smell (CN I), Vision (CN II), Hearing (CN VIII cochlear), Equilibrium (CN VIII vestibular), Taste (CN VII, IX, X minor).
- **Eyeball Movement (3 Nerves):** Oculomotor (CN III - SR, MR, IR, IO), Trochlear (CN IV - SO), Abducens (CN VI - LR).
- **Skeletal Muscle Innervation:** Facial (CN VII - facial expression), Spinal Accessory (CN XI - SCM & trapezius), Trigeminal (CN V - chewing: masseter, temporalis), Vagus (CN X - larynx/speech), Hypoglossal (CN XII - tongue).
- **4 Parasympathetic Cranial Nerves:** CN III (pupil constriction, ciliary focus), CN VII & IX (salivary & tear glands), CN X (visceral organs: heart, lungs, kidneys, liver, stomach, intestines).

Clinical Pathology Correlation:

- **Anosmia (Loss of smell):** Fractures of the cribriform plate damage the Olfactory nerve (CN I) as it passes through.
- **Bell's Palsy:** Unilateral paralysis of the muscles of facial expression caused by inflammation of the Facial nerve (CN VII).
- **Tic Douloureux (Trigeminal Neuralgia):** Severe facial pain caused by inflammation of the Trigeminal nerve (CN V).
- **Blindness:** Optic nerve (CN II) damage; degree of blindness depends on exact lesion location.

MODULE 3: Sympathetic vs. Parasympathetic Functional Architecture

The ANS maintains internal homeostasis through dual innervation of most visceral organs, balancing sympathetic ('fight or flight') and parasympathetic ('rest and digest') activities.

Comparison of Somatic & Autonomic Efferent Pathways

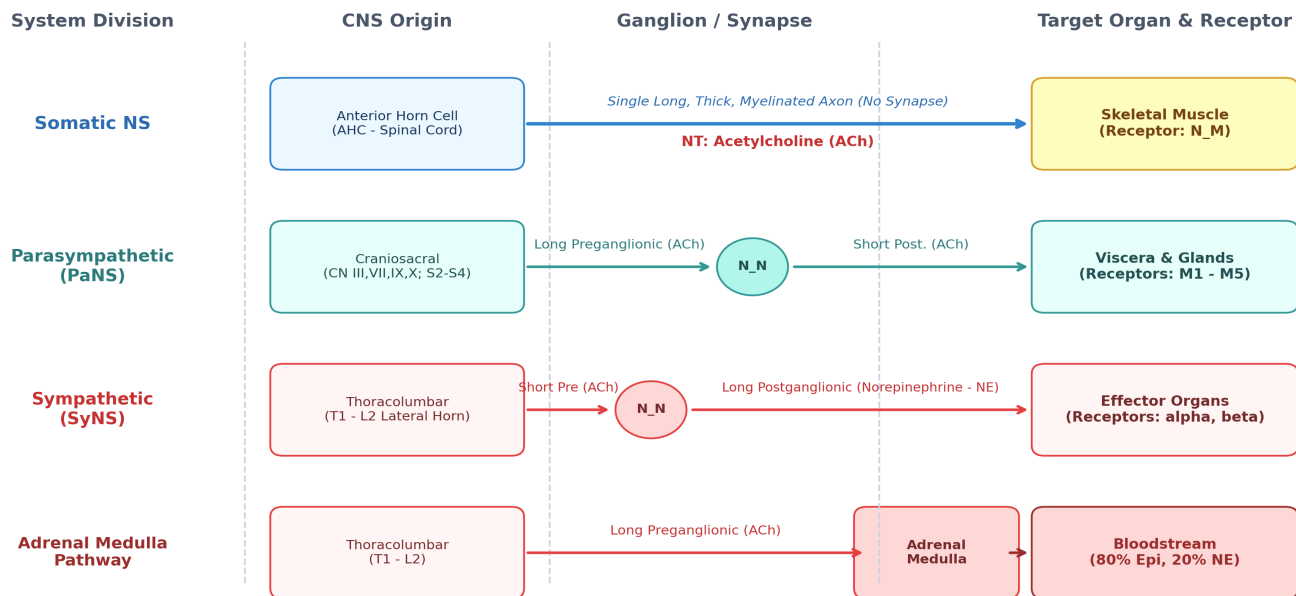


Figure 3.1: Structural Comparison of Somatic, Parasympathetic, Sympathetic, and Adrenal Efferent Pathways. Note differences in preganglionic axon length, ganglion location, and neurotransmitter release.

1. Sympathetic Nervous System (SyNS) — Thoracolumbar Outflow:

- **Origin & Wiring:** Cell bodies of preganglionic motor neurons originate in the **lateral horns of T1–T12 and L1–L2** spinal cord segments. Preganglionic axons synapse in lateral (paravertebral chain) or collateral (prevertebral: celiac, superior mesenteric, inferior mesenteric) ganglia far away from target organs. Thus, SyNS contains **short preganglionic neurons** and **long postganglionic neurons**.
- **Physiological Functions:** Mediates 'fight or flight' responses. Operates in conjunction with the adrenal medulla (stimulating release of epinephrine/norepinephrine into blood). Increases Metabolic Rate (MR) and Cardiac Output (CO) during exercise and excitement. Generally **excitatory** to almost all body tissues *except the gastrointestinal tract (GIT)*, where it exerts **inhibitory** effects (decreases motility/secretion, contracts sphincters).

2. Parasympathetic Nervous System (PaNS) — Craniosacral Outflow:

- **Origin & Wiring:** Preganglionic motor cell bodies reside in nuclei of **cranial nerves (CN III, VII, IX, X)** and sacral spinal cord segments (**S2–S4**). Preganglionic axons synapse in **terminal ganglia** located close to or within the target organ wall. Thus, PaNS contains **long preganglionic neurons** and **short postganglionic cholinergic neurons**.
- **Physiological Functions:** Mediates 'rest and digest' activities (regulates digestion, defecation, micturition). Conserves energy, promotes anabolic states, maintains homeostatic baseline. Operates via discrete localized actions (no mass discharge) with continuous tonic discharge to the heart (vagal tone). Generally **inhibitory** to most tissues *except the GIT*, where it exerts **excitatory** effects (increases motility/secretion, relaxes sphincters).

EXAM ESSENTIAL: Structures Lacking Dual Innervation

Dual Innervation & Unique Exceptions:

Most body structures receive dual innervation by both PaNS and SyNS for precise balance and control.

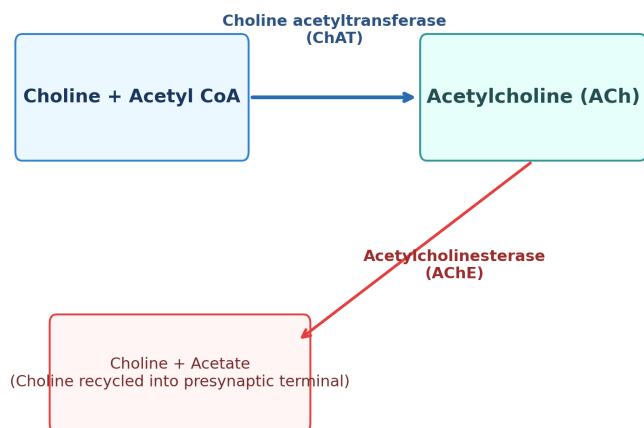
Structures that DO NOT receive dual innervation (Innervated ONLY by the Sympathetic System):

1. **Sweat glands** (Sympathetic cholinergic postganglionic innervation)
2. **Adrenal medulla** (Innervated directly by sympathetic preganglionic fibers)
3. **Blood vessels** (Vascular smooth muscle tone controlled by sympathetic tone)
4. **Skin piloerectors** (Arrector pili muscles causing hair erection)

MODULE 4: Neurotransmitters, Receptors & Second Messenger Cascades

The ANS utilizes two main neurotransmitters: **Acetylcholine (ACh)** and **Norepinephrine (NE)**. All preganglionic neurons (SyNS and PaNS) and all parasympathetic postganglionic neurons release ACh (cholinergic). Most sympathetic postganglionic neurons release NE (adrenergic), with exceptions such as sweat glands.

A. Acetylcholine (ACh) Synthesis & Degradation



B. Norepinephrine (NE) Biosynthetic Pathway

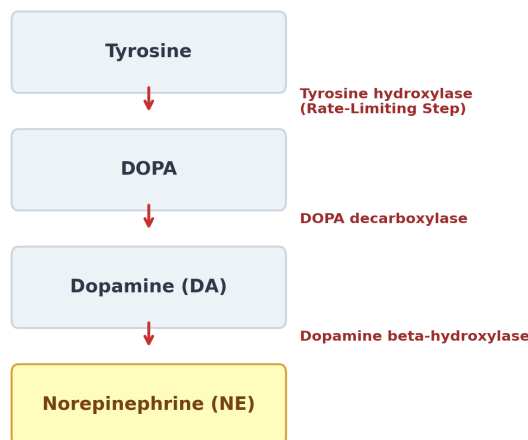


Figure 4.1: Biosynthetic & Enzymatic Pathways for Acetylcholine and Norepinephrine. Note Tyrosine Hydroxylase as the rate-limiting enzyme in catecholamine synthesis.

Neurotransmitter Synthesis Pathways:

- **Acetylcholine Synthesis:** Formed from Choline and Acetyl CoA via the enzyme **Choline acetyltransferase (ChAT)**: *Choline + Acetyl CoA* → *ACh + CoA*. Degraded rapidly in the synaptic cleft by **Acetylcholinesterase (AChE)** into choline (recycled) and acetate.
- **Norepinephrine Biosynthesis:** Sequential enzymatic conversion: *Tyrosine* → (Tyrosine hydroxylase) → *DOPA* → (DOPA decarboxylase) → *Dopamine (DA)* → (Dopamine β-hydroxylase) → *Norepinephrine (NE)*.

Autonomic Receptors & Signal Transduction Second Messenger Systems

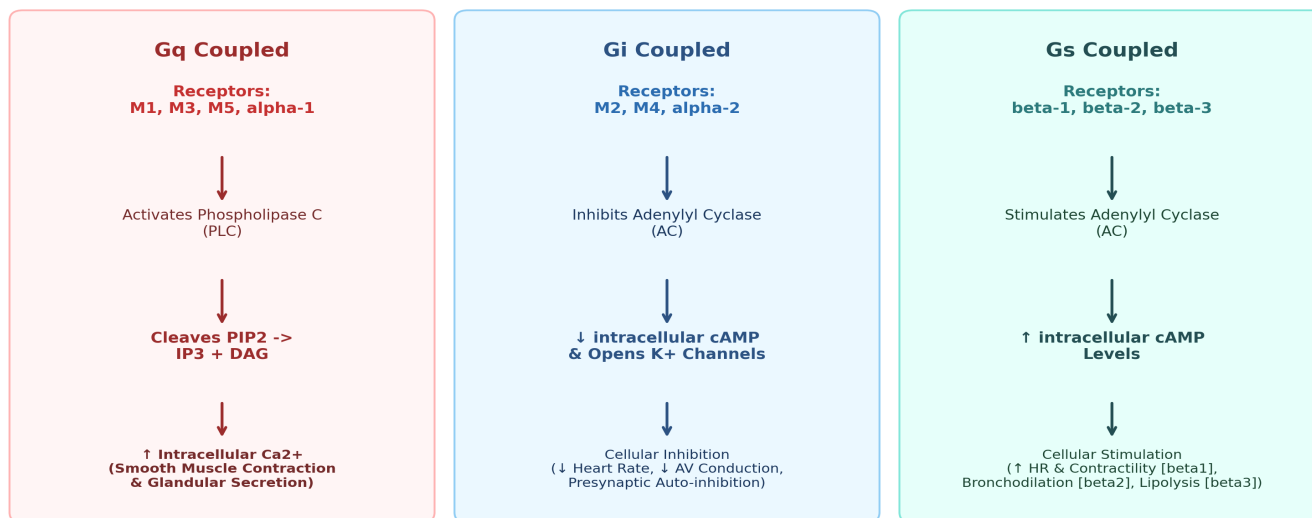


Figure 4.2: Autonomic Receptor Subtypes & Second Messenger Transduction Mechanisms. Gq increases Ca²⁺; Gi decreases cAMP; Gs increases cAMP.

Autonomic Receptor Subtypes & Intracellular Mechanisms:

Receptor Family	Subtype	Primary Locations	Intracellular Transduction & Second Messenger Result
Cholinoceptors	Muscarinic M1	CNS neurons, sympathetic postganglionic neurons	Gq coupled → Formation of IP3 and DAG → ↑ intracellular Ca ²⁺
Cholinoceptors	Muscarinic M2	Myocardium (heart), smooth muscle, CNS neurons	Gi coupled → Opening of K ⁺ channels, inhibition of adenylyl cyclase (↓ cAMP)
Cholinoceptors	Muscarinic M3	Exocrine glands, vessels, bronchial smooth muscle, GIT	Gq coupled → Formation of IP3 and DAG → ↑ intracellular Ca ²⁺
Cholinoceptors	Muscarinic M4	CNS neurons, vagal nerve endings	Gi coupled → Inhibition of adenylyl cyclase (↓ cAMP)
Cholinoceptors	Muscarinic M5	Vascular endothelium (cerebral vessels), CNS neurons	Gq coupled → Formation of IP3 and DAG → ↑ intracellular Ca ²⁺
Cholinoceptors	Nicotinic N _N	Postganglionic neurons (autonomic ganglia), adrenal medulla	Ligand-gated ion channel → Opening of Na ⁺ /K ⁺ channels → Depolarization
Adrenoceptors	Alpha-1 (α ₁)	Postsynaptic effector cells, especially smooth muscle	Gq coupled → Formation of IP3 and DAG → ↑ intracellular Ca ²⁺
Adrenoceptors	Alpha-2 (α ₂)	Presynaptic adrenergic terminals, platelets, lipocytes, smooth muscle	Gi coupled → Inhibition of adenylyl cyclase → ↓ intracellular cAMP
Adrenoceptors	Beta-1 (β ₁)	Postsynaptic effector cells (Heart, lipocytes, brain)	Gs coupled → Stimulation of adenylyl cyclase → ↑ intracellular cAMP
Adrenoceptors	Beta-2 (β ₂)	Smooth muscle (bronchial, vascular, GI), cardiac muscle	Gs coupled → Stimulation of adenylyl cyclase → ↑ intracellular cAMP
Adrenoceptors	Beta-3 (β ₃)	Postsynaptic effector cells (Lipocytes, heart)	Gs coupled → Stimulation of adenylyl cyclase → ↑ intracellular cAMP

MODULE 5: Systemic Organ Effects Matrix

The physiological actions of sympathetic and parasympathetic stimulation across major organ systems are mediated by specific receptor distributions:

Effector Organ / System	Sympathetic Receptor	Sympathetic Effect	Parasympathetic Effect (M3/M2)
Eye: Iris Radial Muscle	Alpha-1	Contracts (Pupil dilation / Mydriasis)	—
Eye: Iris Sphincter Muscle	—	—	Contracts (Pupil constriction / Miosis)
Eye: Ciliary Muscle	Beta	Relaxes (Flattens lens for far vision)	Contracts (Convex lens for near vision / accommodation)
Heart: SA Node	Beta-1	Increases Heart Rate (Positive Chronotropy)	Decreases Heart Rate (Negative Chronotropy - M2)
Heart: Atria & AV Node	Beta-1	Increases Contractility & Conduction Velocity	Decreases Contractility & Conduction Velocity (M2)
Heart: Ventricles	Beta-1	Increases Contractility	Decreases contractility slightly (M2)
Arterioles: Coronary	Alpha / Beta	Constricts (Alpha) / Dilates (Beta)	—
Arterioles: Skin & Viscera	Alpha	Constricts	Dilates (Salivary glands only)
Arterioles: Skeletal Muscle	Alpha / Beta-2	Constricts (Alpha) / Dilates (Beta-2)	—
Lungs: Bronchial Smooth Muscle	Beta-2	Relaxes (Bronchodilation)	Contracts (Bronchoconstriction - M3)
Lungs: Bronchial Glands	Alpha / Beta	Inhibits (Alpha) / Stimulates (Beta)	Stimulates secretion (M3)
Stomach & Intestines: Motility	Alpha / Beta	Decreases Motility and Tone	Increases Motility and Tone (M3)

Stomach & Intestines: Sphincters	Alpha-1	Contracts Sphincters	Relaxes Sphincters (M3)
Stomach & Intestines: Secretion	Alpha	Inhibits Secretion	Stimulates Secretion (M3)
Gallbladder	Beta-2	Relaxes Gallbladder	Contracts Gallbladder
Liver	Alpha / Beta-2	Glycogenolysis & Gluconeogenesis (↑ Blood Glucose)	—
Pancreas: Exocrine & Endocrine	Alpha / Beta	Inhibits Exocrine/Insulin (Alpha); Stimulates Insulin (Beta)	Stimulates Exocrine Secretion
Urinary Bladder: Detrusor Wall	Beta-2	Relaxes Bladder Wall	Contracts Bladder Wall (M3 - Micturition)
Urinary Bladder: Sphincter	Alpha-1	Contracts Internal Sphincter	Relaxes Internal Sphincter (M3)
Male Reproductive Tract	Alpha-1	Ejaculation ('Shoot' - Sympathetic)	Erection ('Point' - Parasympathetic)
Skin: Sweat Glands & Piloerectors	Alpha / Cholinergic M	Localized secretion (Alpha); Generalized secretion (M); Hair erection	—
Kidneys & Adipocytes	Beta-1 / Beta-3	Increases Renin Secretion (Beta-1); Increases Fat Breakdown / Lipolysis (Beta-3)	—

MODULE 6: Autonomic Reflexes & Reflex Arc Architecture

An **autonomic reflex** is a fast, involuntary response to a physiological stimulus. Every autonomic reflex action is transmitted through a **reflex arc** consisting of 5 essential components:

1. **Receptor:** Detects internal/external environment changes and undertakes signal transduction.
2. **Afferent (Sensory) Pathway:** Conducts sensory action potentials from the receptor to the CNS.
3. **Integrating Center:** Regions in the CNS (Spinal cord, brain stem, hypothalamus, cerebral cortex) that process sensory input and determine the reflex response.
4. **Efferent (Motor) Pathway:** Conducts motor action potentials from the CNS to effector organs via pre- and postganglionic autonomic fibers.
5. **Effector Organs:** Visceral target tissues (Cardiac muscle, smooth muscle, glands) that receive feedback and respond accordingly.

AUTONOMIC REFLEX SUMMARY

Representative Autonomic Reflexes:

- **Baroreceptor Reflex:** Rapid regulation of arterial blood pressure via carotid sinus and aortic arch stretch receptors.
- **Chemoreceptor Reflex:** Monitoring and regulation of blood gases (PaO₂, PaCO₂, pH).
- **Defecation Reflex:** Involuntary rectal smooth muscle contraction and anal sphincter relaxation.
- **Micturition Reflex:** Contraction of detrusor muscle and relaxation of internal urinary sphincter.
- **Visceral Motor Reflexes:** Regulation of heart rate, alteration of respiratory depth/rate, digestive motility, and pupil diameter.

MODULE 7: Pharmacology of the Autonomic Nervous System

Pharmacological agents manipulate autonomic neurotransmission by acting as direct receptor agonists, indirect enhancers, synthesis inhibitors, release blockers, or direct receptor antagonists.

Drug Category	Pharmacological Mechanism	Specific Representative Drugs
Direct Sympathomimetics	Directly stimulate adrenergic alpha and/or beta receptors.	Epinephrine, Norepinephrine, Methoxamine
Selective Adrenergic Agonists	Stimulate specific adrenoceptor subtypes preferentially.	Phenylephrine (alpha-1 selective), Isoproterenol (non-selective beta), Albuterol (beta-2 selective)
Indirect Sympathomimetics	Promote release of catecholamines from presynaptic adrenergic nerve terminals.	Ephedrine, Tyramine, Amphetamine, Cathinone / Cathin

Inhibitors of NE Synthesis & Storage	Disrupt vesicular storage and synthesis of norepinephrine.	Reserpine
Adrenergic Release Blockers	Inhibit release of norepinephrine from presynaptic nerve terminals.	Guanethidine
Alpha-Adrenergic Blockers	Block postsynaptic alpha-1 and alpha-2 receptors.	Phenoxybenzamine (irreversible), Phentolamine (reversible)
Beta-Adrenergic Blockers	Block beta-1 and/or beta-2 adrenoceptors.	Propranolol (non-selective beta-1/beta-2), Metoprolol (beta-1 selective)
Ganglionic Blockers	Block nerve impulse transmission through autonomic ganglia (NN) of both SyNS and PaNS.	Hexamethonium
Direct Parasympathomimetics	Directly stimulate muscarinic cholinergic receptors.	Pilocarpine, Methacholine
Anticholinesterases (Indirect PaNS)	Inhibit acetylcholinesterase enzyme, potentiating ACh duration and action.	Neostigmine, Physostigmine, Ambenonium, Malathion (organophosphate)
Direct Muscarinic Blockers	Competitively block muscarinic receptors at effector organs.	Atropine, Homatropine, Hyoscine / Scopolamine
Direct Nicotinic Blockers	Block nicotinic neuromuscular/ganglionic receptors.	Curare
Indirect Cholinergic Toxins	Inhibit ACh release OR cause massive release followed by vesicle depletion.	Botulinum Toxin (inhibits ACh release); Black Widow Spider Venom (massive ACh release/depletion)
Postganglionic Ganglionic Stimulators	Stimulate autonomic postganglionic neuronal nicotinic receptors.	Acetylcholine, Nicotine

MODULE 8: High-Yield Conclusion & Exam Summary Note

EXAM SUMMARY NOTE: RAPID RECALL POINTS

Top 12 High-Yield Exam Review Points for Rapid Recall:

- Outflow Anatomy:** Sympathetic = Thoracolumbar (T1–L2 lateral horn); Parasympathetic = Craniosacral (CN III, VII, IX, X & S2–S4).
- Neuron Lengths:** Sympathetic = Short preganglionic, Long postganglionic; Parasympathetic = Long preganglionic, Short postganglionic.
- Preganglionic NT:** ALL preganglionic neurons (SyNS and PaNS) release Acetylcholine (ACh) onto Nicotinic (NN) receptors.
- Single Innervation Exceptions:** Sweat glands, Adrenal medulla, Blood vessels, and Piloerectors receive ONLY sympathetic innervation.
- Catecholamine Rate-Limiting Enzyme:** **Tyrosine Hydroxylase** converts Tyrosine to DOPA.
- G-Protein Signalling Cascades:**
 - Gq (IP3/DAG → ↑ Ca²⁺):** M1, M3, M5, Alpha-1 (HAVE1 M&M's).
 - Gi (↓ cAMP):** M2, M4, Alpha-2 (MAD 2's).
 - Gs (↑ cAMP):** Beta-1, Beta-2, Beta-3.
- Cranial Nerves Parasympathetic Outflow:** CN III (miosis/lens focus), CN VII (lacrimal/salivary), CN IX (salivary), CN X (thoracoabdominal viscera).
- Cranial Pathologies:** CN I cribriform fracture = Anosmia; CN VII inflammation = Bell's Palsy; CN V inflammation = Tic Douloureux.
- Organ Effects Rule:** Sympathetic relaxes bronchial muscle (β_2), contracts sphincters (α_1), increases HR (β_1), inhibits GI motility. Parasympathetic contracts bronchial muscle (M3), relaxes sphincters (M3), decreases HR (M2), stimulates GI motility.
- Ganglionic Blocker:** **Hexamethonium** blocks transmission across autonomic ganglia of both SyNS and PaNS.
- Adrenergic Synthesis Blocker:** **Reserpine** prevents NE synthesis/storage; **Guanethidine** blocks NE release.
- Toxins:** **Botulinum Toxin** blocks ACh release (flaccid paralysis); **Black Widow Spider Venom** causes massive ACh release and depletion.

MODULE 9: High-Yield Multiple Choice Questions (MCQs) & Answers

Q1. A 45-year-old male presents with severe facial pain triggered by light touching of his cheek. The attending physician diagnoses Tic Douloureux (Trigeminal Neuralgia). Which cranial nerve is inflamed?

- Cranial Nerve III
- Cranial Nerve V
- Cranial Nerve VII

D. Cranial Nerve IX

Correct Answer: B. Cranial Nerve V (Trigeminal Nerve). Explanation: Passage 10 states that Tic Douloureux is severe facial pain caused by inflammation of the Trigeminal nerve (CN V).

Q2. Which structural feature distinguishes the Parasympathetic Nervous System from the Sympathetic Nervous System?

- A. Long preganglionic and short postganglionic neurons
- B. Origin from lateral horns of T1-L2 spinal cord
- C. Short preganglionic and long postganglionic neurons
- D. Synapse in paravertebral sympathetic chain ganglia

Correct Answer: A. Long preganglionic and short postganglionic neurons. Explanation: Passages 13 & 15 detail that parasympathetic neurons contain long preganglionic fibers synapsing in terminal ganglia near organs, whereas sympathetic neurons have short preganglionic fibers.

Q3. Which of the following organs or tissues receives ONLY sympathetic innervation (lacks dual autonomic innervation)?

- A. Urinary bladder detrusor muscle
- B. Heart SA node
- C. Sweat glands
- D. Iris sphincter muscle

Correct Answer: C. Sweat glands. Explanation: Passage 12 explicitly lists sweat glands, adrenal medulla, blood vessels, and skin piloerectors as structures that do not receive dual innervation.

Q4. What is the rate-limiting enzyme in the synthesis pathway of Norepinephrine from Tyrosine?

- A. DOPA decarboxylase
- B. Choline acetyltransferase
- C. Tyrosine hydroxylase
- D. Dopamine beta-hydroxylase

Correct Answer: C. Tyrosine hydroxylase. Explanation: Passage 24 illustrates the norepinephrine synthesis pathway starting with Tyrosine hydroxylase converting Tyrosine to DOPA.

Q5. Stimulation of Muscarinic M3 receptors on bronchial smooth muscle leads to bronchoconstriction through which second messenger cascade?

- A. Inhibition of adenylyl cyclase and decreased cAMP
- B. Activation of Phospholipase C, yielding IP3 and DAG to increase intracellular Ca²⁺
- C. Direct opening of nicotinic sodium channels
- D. Stimulation of adenylyl cyclase and increased cAMP

Correct Answer: B. Activation of Phospholipase C, yielding IP3 and DAG to increase intracellular Ca²⁺. Explanation: Passages 21 & 30 state that M3 receptor activation stimulates Phospholipase-C, producing IP3 + DAG and increasing intracellular Ca²⁺ influx.

Q6. A patient presenting with Bell's Palsy exhibits unilateral paralysis of the muscles of facial expression. Which nerve is affected?

- A. Facial Nerve (CN VII)
- B. Hypoglossal Nerve (CN XII)
- C. Spinal Accessory Nerve (CN XI)
- D. Trigeminal Nerve (CN V)

Correct Answer: A. Facial Nerve (CN VII). Explanation: Passage 10 notes that Bell's Palsy is paralysis of facial expression muscles caused by inflammation of the Facial nerve (CN VII).

Q7. Which pharmacological agent acts as a ganglionic blocker by preventing impulse transmission through autonomic ganglia of both PaNS and SyNS?

- A. Propranolol
- B. Atropine
- C. Hexamethonium
- D. Neostigmine

Correct Answer: C. Hexamethonium. Explanation: Passage 28 lists Hexamethonium under drugs blocking nerve impulse transmission through autonomic ganglia of both PaNS and SyNS.

Q8. Which cranial nerve provides parasympathetic innervation to major visceral organs including the heart, lungs, stomach, and intestines?

- A. Oculomotor Nerve (CN III)
- B. Glossopharyngeal Nerve (CN IX)
- C. Vagus Nerve (CN X)
- D. Facial Nerve (CN VII)

Correct Answer: C. Vagus Nerve (CN X). Explanation: Passages 6 & 9 highlight the Vagus nerve (CN X) as the major nerve of the parasympathetic system going to heart, lungs, kidneys, liver, stomach, and intestines.

Q9. What is the effect of Sympathetic stimulation on the Urinary Bladder detrusor wall and internal sphincter?

- A. Contracts detrusor wall (M3) and relaxes sphincter
- B. Relaxes detrusor wall (Beta-2) and contracts internal sphincter (Alpha-1)
- C. Contracts detrusor wall and contracts sphincter
- D. Relaxes detrusor wall and relaxes sphincter

Correct Answer: B. Relaxes detrusor wall (Beta-2) and contracts internal sphincter (Alpha-1). Explanation: Passage 20 details that sympathetic activity relaxes the bladder wall via Beta receptors and contracts the internal sphincter via Alpha receptors to promote urinary retention.

Q10. Which toxin acts as an indirect cholinergic antagonist by inhibiting the presynaptic release of Acetylcholine?

- A. Black Widow Spider Venom
- B. Botulinum Toxin
- C. Muscarine
- D. Scopolamine

Correct Answer: B. Botulinum Toxin. Explanation: Passage 30 lists Botulinum Toxin under indirect cholinergic antagonists that block ACh release.